Group 5: Gamers

A Comparative Analysis of Critic and Player Reviews in Video Games

CS-133 Section 01 - Data Visualization

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Abstract

Nowadays, the video game industry depends a lot on rating systems to help players decide whether they should buy or try a game. However, in reality, critics' scores do not always match players' opinions. Critics often focus on things such as graphics, technical quality, or creativity, while players usually care more about gameplay and the overall entertainment value. Because of that, there are clear differences between these two perspectives. This raises the question of whether critics' reviews really reflect the actual experience of players.

In this project, we use the Steam Games Dataset (March 2025) from Kaggle, which contains more than 90,000 games, including Metacritic scores, player reviews, peak player count, and game descriptors such as genres and developer-written descriptions. We perform exploratory data analysis (EDA) to examine patterns between critic and player ratings and use visualizations such as bar charts and box plots to identify relationships between key variables. We then apply machine learning models, including logistic regression and ridge regression, along with TF-IDF (Term Frequency Inverse Document Frequency), to analyze game descriptions and explore factors that may influence player opinions.

Our analysis shows that critic scores and player ratings are related, but not consistently. In several cases, especially across certain genres, clear gaps can be seen between the two. We also observe that factors such as gameplay-related features, tags, and player engagement have a stronger influence on player ratings than critic scores alone.

Overall, our findings suggest that critic ratings cannot fully represent player experiences, and that player reviews are still important for understanding how games are actually experienced.

Background & Problem Definition

This project uses a dataset sourced from Kaggle, specifically the Steam Games Dataset (March 2025) compiled by the authors indicated by their account names: Ermilov, Shekmanish, and Gupta. The dataset contains 89,618 video game entries distributed across 47 features, showing both numerical and text-based information. Each row represents a single game and includes identifiers such as *appid* and *name*, metadata like *release_date*, *price*, and *required_age*, and engagement-related metrics such as *peak_ccu* (peak concurrent users), *average_playtime*, and *num_reviews_total*.

A good aspect of the dataset is its combination of critic and player perspectives. Critic opinion is represented through the attribute of *metacritic_score*, while player sentiment is reflected in attributes such as *positive*, *negative*, *pct_pos_total*, and *pct_pos_recent*. Additionally, the dataset includes descriptive text fields (*short_description*, *about_the_game*) and categorical attributes (*genres*, *tags*, *developers*, *publishers*), allowing both structured and unstructured analysis. The dataset, however, also requires clean-up due to missing values in fields like *reviews*, *metacritic_url*, and *score_rank*, which introduces the need for preprocessing. This dataset was selected because it captures both quantitative performance metrics and qualitative descriptors of games, making it optimal for exploring patterns in public and critical reception.

The main problem explored in this project is the potential discrepancy between critic reviews and player reviews in video games. While one might expect alignment between professional evaluations and player experiences, anecdotal evidence from what we've seen in the media suggests that this is not always the case. Critics may emphasize technical quality, innovation, or artistic direction, whereas players may prioritize enjoyment, gameplay mechanics, and long-term engagement. This disconnect raises important questions about bias, influence, and the reliability of review systems, particularly in an industry where consumer decisions are heavily shaped by ratings and visibility.

To guide our analysis, the following five analytical questions were established by our group:

1. Is there a measurable difference between critic reviews and player reviews?
2. Do certain developers or publishers receive systematically higher or lower scores from critics or players?
3. Do AAA titles receive statistically higher ratings compared to smaller or independent games?
4. To what extent do critic scores influence player sentiment and review outcomes?
5. Which game attributes (e.g., genre, tags, keywords, or price) have the strongest impact on public opinion and engagement?

These questions directly inform the types of visualizations and analyses performed during exploratory data analysis. For example, scatter plots comparing *metacritic_score* and *pct_pos_total* help identify correlations or gaps between critic and player sentiment. Bar charts and grouped visualizations of genres and tags reveal which categories are favored differently by critics and players. Heatmaps and correlation matrices provide insight into how variables such

as *price*, *playtime*, and *peak_ccu* relate to review outcomes. These visual insights also act as a basis for our feature selection and hypothesis formation in machine learning models.

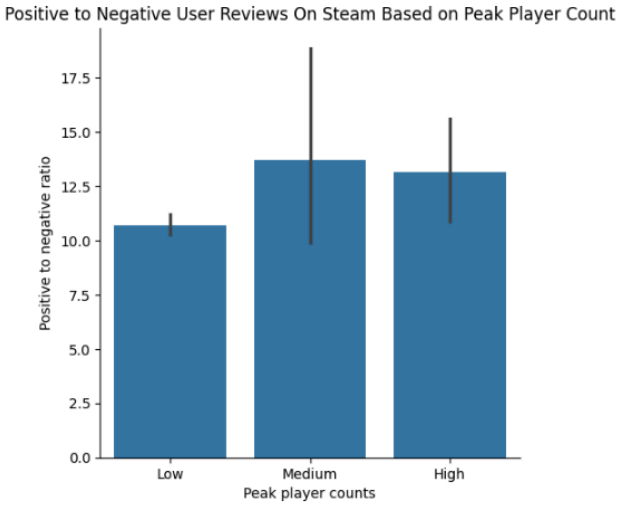
The relevance of this problem goes beyond academic curiosity. The video game industry is a multi-billion-dollar economy where perception can define success or failure for big game development companies and small indie developers. A mismatch between critic and player reviews can influence purchasing decisions, marketing strategies, and even the long-term reputation of developers. Understanding these dynamics allows for more accurate predictive modeling of game success and provides insights into consumer behavior, bias, and trust in review systems. In an area of media where numbers shape narratives, we, as a group, want to find out whether those numbers are truly aligned.

Data Visualization Strategy & Interactive Component

Static Visualizations

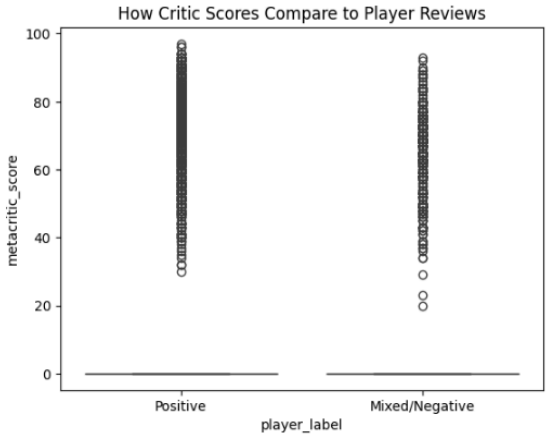
In order to explore the dataset and answer our analytical questions, we made a set of visualizations designed to show the relationships between critic scores, player sentiment, and game characteristics. The following questions were addressed through visualization:

Does a game's peak player count influence its overall critical reception from users?



For this bar chart, we distinguish between low, medium, and high player counts by putting them in bins, where a low peak player count ranges from 50 to 5,000 players, a medium peak player count ranges from 5,000 to 10,000 players, and a high peak player count ranges from 10,000 to 2,500,000 players. We then compare their positive to their negative review ratio, and based on the bar chart, games with peak player counts ranging from 5,000 to 10,000 players tend to have higher positive to negative review ratios (more positive reviews) compared to low or high peak player count games.

Is there a difference between the critic scores and player reviews?



For this visual, we first derived a feature called 'player_ratio' by dividing positive reviews by total reviews to quantify overall player sentiment. We then converted the feature 'player_ratio' into a categorical label ('player_label'), splitting games into *Positive* (>= 0.7) and *Mixed/Negative* (<

0.7). Finally, we used a box plot to compare the distribution of ‘metacritic_score’ across these two groups. Based on the box plot, critic scores are widely spread for both groups, with heavy overlap between them. There is a slight positive correlation between positive player reviews and Metacritic score compared to negative player reviews. However, there is no strong separation between the two groups, meaning there is no significant difference between player and Metacritic reviews, which most likely suggests that other factors are at play.

Interactive Visualizations

In addition to our static plots, we also made interactive visualizations to allow deeper exploration of the dataset. Interactive visualizations allow users to dynamically filter and examine relationships between variables.

To do this, we use Plotly to create interactive scatter plots where users can hover over each point to see more details about a specific game, such as its name, price, and scores. One of our main plots looks at the relationship between peak concurrent users (peak_ccu) and the percentage of positive reviews (pct_pos_total). This makes it easier to connect overall trends with real examples instead of just looking at abstract data points.

We also added a dropdown menu so users can filter the data by genre. This helps break the data into smaller groups and makes it easier to see different critical scores and player opinions depending on the type of game. Without this, it would be harder to notice patterns that only appear within certain genres.

One of our interactive visualizations compares player review ratios and Metacritic scores with the number of discounts. This helps us explore whether discounts have any effect on how games are received. By hovering over each data point, users can see specific game details and identify which games stand out, making the analysis clearer and more engaging.

Overall, these interactive visualizations make the data easier to explore and understand. Instead of looking at fixed graphs, users can interact with the data, which help reveal patterns and relationships more clearly. From what we observed, we decided to use some features such as price, peak player count, number of reviews, and game descriptions for our machine models. This leads to the next step, that we apply machine learning to predict player sentiment.

Machine Learning Pipeline & Evaluation

To better understand the relationship between critic reviews and player opinions, we apply a machine learning model using both numerical and text features.

First, we preprocess the game description by converting all text to lowercase and removing special characters. Then, we use TF-IDF to turn the text into numerical features, so that the model can get important words and phrases from each description.

The logistic regression model also required engineering a new categorical feature from the Metacritic score, so the score was categorized into good, bad, and mixed.

Logistic Regression Model Pipeline:

```
# Pipeline
pipeline = Pipeline([
    ("tfidf", TfidfVectorizer(
        max_features=10000,
        stop_words='english',
        ngram_range=(1, 2),
        min_df=3
    )),
    ("logreg", LogisticRegressionCV(
        Cs=5,
        cv=5,
        max_iter=1000,
        n_jobs=-1,
        class_weight='balanced'
    ))
])
```

LogisticRegressionCV handles some hyperparameter tuning for us by finding an optimal regularization parameter C, and Cs specifies that it will generate 5 different C values across a log scale. This helps to prevent overfitting through regularization by punishing large coefficients. Setting the class_weight to balanced was also crucial because our dataset has a lack of poorly rated games. This somewhat hints that critics such as Metacritic may have some bias, particularly positive bias, when it comes to reviewing games.

Ridge Regression Model Pipeline:

```
# Pipeline
pipeline = Pipeline([
    ("tfidf", TfidfVectorizer(
        max_features=10000,
        stop_words='english',
        ngram_range=(1, 3),
        min_df=3
    )),
    ("ridge", RidgeCV(alphas=[0.1, 1.0, 10.0, 100.0]))
])
```

Ridge regression, as a type of linear regression, uses a term called the L2 regularization term to add a penalty equal to the sum of the squared magnitudes of coefficients, and this helps to reduce overfitting. The alpha controls the strength of the L2 term, so the pipeline performs some hyperparameter tuning for us.

Logistic Regression Price Pipeline:

```
# Pipeline
pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('logreg', LogisticRegression(max_iter=1000)) # Increased max_iter for convergence
])
```

To predict game scores by price, the dataset must be normalized first to minimize the influence of outliers in the dataset. Then, the number of iterations is capped at 1,000, so the model will stop if it does not converge sooner.

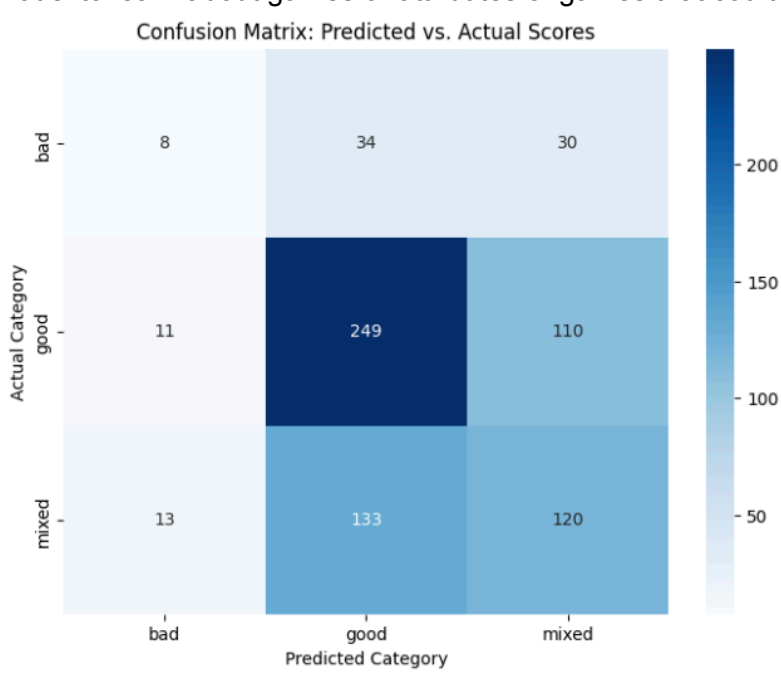
Results, Code Output & Interpretation

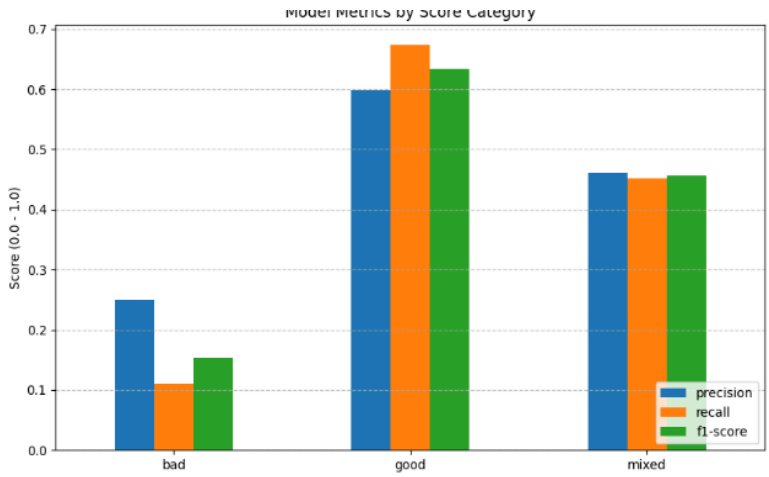
Logistic Regression on Critical Reception:

```
Cross-Validation Accuracy: 0.524 ± 0.019

Test Performance:
Accuracy: 0.532

Classification Report:
              precision    recall  f1-score   support

         bad       0.25      0.11      0.15        72
        good       0.60      0.67      0.63       370
       mixed       0.46      0.45      0.46       266

    accuracy                           0.53       708
   macro avg       0.44      0.41      0.41       708
weighted avg       0.51      0.53      0.52       708
```

The model performs by far the best on games considered good and performs the worst on games considered poor. Similar precision and f1-scores for each category reinforce this interpretation. This highlights the lack of data on poorly rated games in our dataset and shows how data or category bias can hurt the performance of models.

However, this also reveals some insight into the theory that critics may have some positivity bias, because there appears to be a general lack of bad reviews. This makes it hard for our model to learn about games or attributes of games that could make them be received poorly.





Ridge Regression:

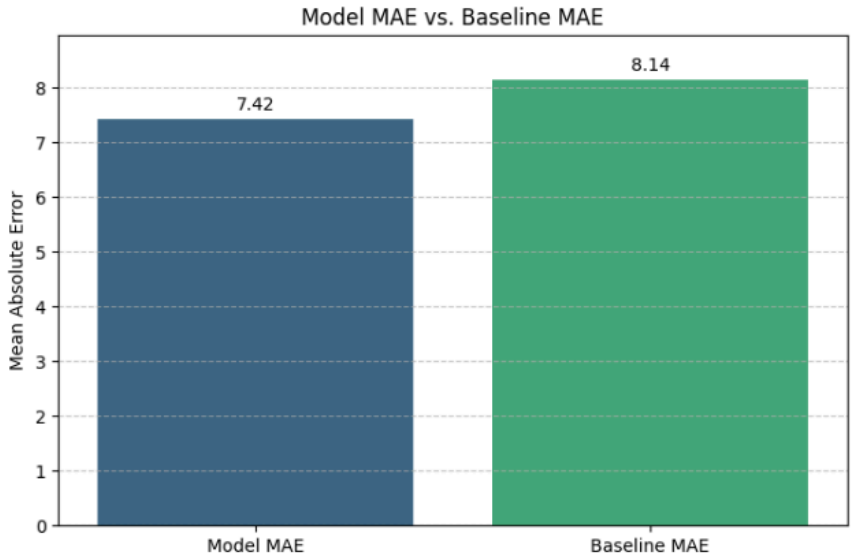
```
Cross-Validation MAE: 7.50 ± 0.37

Final Test Performance:
Test MAE: 7.42
Test R^2: 0.1509
Baseline MAE: 8.14

Top 10 Terms Associated with HIGHER Scores:
             term  coefficient
763        awards    10.910201
7033  puzzle game    10.882150
6633         play    10.311037
2794      edition    10.060718
5718       monkey     9.967705
1728    community     9.483650
5928          new     9.475105
1209        built     9.195115
2093       custom     9.044750
7961         sims     8.958891

Top 10 Terms Associated with LOWER Scores:
            term  coefficient
5639    missions   -15.586086
2435   different   -15.257147
8189     special   -13.378267
3365    features   -13.159538
6368  painkiller   -12.650135
9930       xenos   -11.677991
8336     stealth   -11.240237
9991      zombie   -10.822377
4377      hockey   -10.599810
7875     shelter   -10.581243
```

These metrics indicate that, on average, our model predicts Metacritic scores within about 7.4 points, and is better than simply guessing the average Metacritic score every time, which is 8.14. This means that, while it is very challenging to capture how good a game is based purely on a fairly short word description, our model did indeed learn and manage to perform decently. Our R-squared value is low, but that is pretty expected because words alone or metrics in

general are hard to use to predict how good a game is, because so many factors, like game mechanics, influence how good a game feels and reviews.

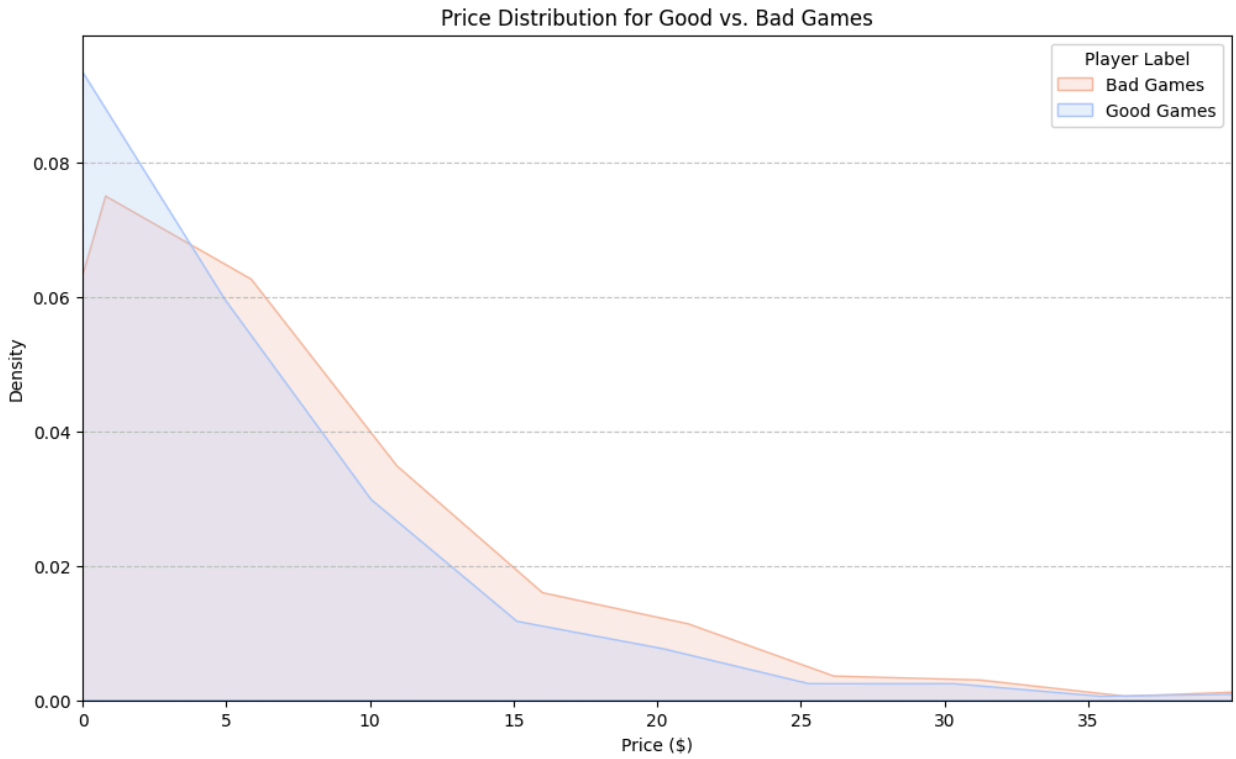
Here we can also see that specific descriptions such as "puzzle game" or "community" correlate to higher scores, so players probably like puzzle games and community support. Terms like "edition" and "awards" likely sway player opinion by suggesting the game is good enough to have multiple editions and awards. "New" is also present, so players like to see innovation.

On the other hand, generic terms like "missions" or "special" or even "zombies" seem to hurt scores. These generic markers seem to potentially turn players away from games, as they may be perceived as overpromising or generic.



Logistic Regression Price:

```
Model Accuracy: 55%
Model F1-Score: 71%
True Positives (TP): 9788
True Negatives (TN): 1
False Positives (FP): 8094
False Negatives (FN): 2
```



According to the distribution of games that are reviewed well and poorly, it appears that the price of these games does not have much influence on their reception. The plot shows that good and bad games tend to overlap with each other, so it is difficult to truly classify a game's reception solely based on its price.

After training the Logistic Regression model, the claim appears to hold true, as the number of true positive games and the number of false positive games appear to be similar, though the number of true positive games is slightly higher. As a result, the model was unable to classify the bad games based on their price, as the number of true negative games and false negative games is much lower than the positive ones.

Reproducibility, GitHub, & Documentation

Our project is fully reproducible through a structured GitHub repository containing all relevant notebooks, code, and supporting files. Because of dataset size constraints, the dataset is not stored directly in the repository. However, it is sourced from Kaggle and loaded through Google Drive within Google Colab.

The repository is organized into all relevant notebooks for exploratory data analysis (EDA) and preprocessing within the 'Project Assignments' folder, ML models in a separate folder, this project report file, and a README file.

To ensure reproducibility, users will be instructed in the README file to download the Kaggle dataset, upload it to their Google Drive, and update the file paths at the beginning of each Colab notebook, if need be, before running the notebooks. The environment is based on Google Colab, which minimizes setup requirements for the user. Due to this, required libraries that the user would need (pandas, numpy, scikit-learn, matplotlib, seaborn, etc.) are imported within each notebook, and users would need no further actions in order to get the Colab notebooks working.

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